

QMS 101 Introductory Statistics

STATISTICAL DATA PORTFOLIO ASSIGNMENT

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1. Introduction

This assignment is designed to help students apply statistical concepts learned in QMS 101 in real-life situations. Students will collect, analyze, and interpret data step-by-step and present their findings. Topics will be provided by the lecturer.

2. Objectives

General Objectives

Students will:

- ▶ Collect and organize real-world data
- ▶ Present data using appropriate graphs
- ▶ Apply statistical methods (mean, median, mode, etc.)
- ▶ Interpret data for decision-making
- ▶ Develop teamwork and presentation skills

Learning Outcomes

By the end of the assignment, students should be able to:

- ▶ Understand real-world data collection
- ▶ Apply statistical methods practically
- ▶ Interpret business and social data
- ▶ Work effectively in teams
- ▶ Communicate findings clearly

3. Group Formation

- ▶ Students will work in groups of **10 members**
- ▶ Groups should be from the **same or related programs**
- ▶ Each group will work on **ONLY ONE assigned topic**
- ▶ Deadline for Submitting Groups: **17 April 2026**

Task 1: Data Collection Tool Design

- ▶ Design **questionnaire/data sheet/Observation check list**
 - ▶ Check the lecture slides (designing a questionnaire)
 - ▶ **Not more than 10 questions**
- ▶ Pilot test (**5 respondents**)
- ▶ Revise based on feedback

Important Notes

▶ **Deadline**

- ▶ Submit your tool by **24 April 2026 before 8.00 AM** via anna.fome@juco.ac.tz
- ▶ Late submission will attract a penalty
- ▶ **“WRITE ONLY THE GROUP MEMBERS WHO PARTICIPATED IN THAT TASK”**

Important Notes...

- ▶ How?
 - ▶ At this stage: Online submission only
 - ▶ Do not print any document
- ▶ Next Task
 - ▶ To be Communicated

Task 2: Data Collection

- ▶ Administer the questionnaire or any tool your group prepared (paper or digital) to collect responses.
- ▶ **Output:** Cleaned raw dataset/table in **Excel format (.xlsx preferred)**.
- ▶ **Submit to:** anna.fome@juco.ac.tz
- ▶ **Deadline:** 01.05.2026
- ▶ **File name format:** **GROUPCODE_Task2.xlsx**
- ▶ **Example:** BP&BE4_Task2.xlsx
- ▶ Late submission will attract a penalty.
- ▶ Write **only names of members who participated in this task**.

Data Collection (Low or No Cost)

Use affordable and accessible methods such as:

- ▶ Asking classmates and friends
- ▶ Free online tools (e.g., Google Forms)
- ▶ Face-to-face data collection on campus
- ▶ WhatsApp, email, or social media platforms

Task 3: Frequency Distribution Table

- ▶ From the data you collected select one appropriate variable that can be used to calculate the mean, median and mode.
- ▶ Construct one grouped frequency distribution table containing all necessary columns required for the calculation of mean, median and mode

Task 4: Displaying data

Plot 1: Histogram and Frequency Polygon

- ▶ Using the data from Task 3, construct: a **histogram**, and a **frequency polygon** on the **same x–y axes**.

Plot 2: Ogive and Estimation of Median

- ▶ Construct an **ogive (cumulative frequency curve)** using the cumulative frequencies from the grouped frequency table.
Using the ogive:
 - ▶ estimate the **median** graphically,
 - ▶ and clearly indicate the median point on the graph.

Task 5: Calculations and Interpretation

- ▶ Using the grouped frequency distribution table constructed in Task 3:
 1. Calculate the mean, median, and mode using the appropriate **grouped-data formulas**:
 2. Interpret your findings by explaining what the calculated: mean, median, and mode indicate.
 3. Discuss the **skewness** of the distribution by comparing the values of the mean, median, and mode.

Important Instructions

- ▶ Write these **by hand**.
- ▶ Show **all calculations and working steps** clearly.
- ▶ Submit **before 15 May 2026**.
- ▶ Submission venue: **Office No. 34, Block E**.
- ▶ **Late submission will not be considered.**
- ▶ Do not forget to write:
 - ▶ your **group code**, and
 - ▶ the **names of all participating students**.

